

AUTONOMOUS ML RESEARCH / TECHJAM 2026

# SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous by default. [Supervisable by design.](#)

ROBUST

SUPERVISABLE

BROAD SEARCH

# The run, in four moves

A single route from the problem to the proof.

01



PROBLEM

## Why current agents fail

Set the task and its failure modes.

02



SYSTEM

## A system for sustained research

Keep experience while resetting the scientist.

03



USER EXPERIENCE

## Run the research journey

Move from task contract to retained state.

04



EVALUATION

## From claim to evidence

Measure what survives the run.

# Automating ML experimentation

Given a dataset and a scoring metric, an agent must build a model and improve it through repeated experiments.

|                                                                                                      |                                      |                                        |                                     |                                      |
|------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|--------------------------------------|
| 01                                                                                                   | 02                                   | 03                                     | 04                                  | 05                                   |
| <b>Read the problem</b>                                                                              | <b>Inspect data</b>                  | <b>Engineer features</b>               | <b>Train + tune</b>                 | <b>Evaluate</b>                      |
| What should the model predict?                                                                       | What patterns does the data contain? | Turn raw data into useful model inputs | Fit a model and adjust its settings | Check performance on validation data |
| REFLECT + REVISE                                                                                     |                                      |                                        |                                     |                                      |
| The agent uses the results to choose what to change, revises the code, and runs the next experiment. |                                      |                                        |                                     |                                      |

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## THE AGENT'S JOB

**Write the code and run the full loop autonomously.**

This includes building the pipeline and deciding which experiments to try next.

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## THE CHALLENGE GOAL

**Reproduce, then improve on the organizer's baseline.**

Develop with training data and public validation. The organizer scores the final submission on a hidden test set.

# Why current agents fail

## 01 WORKFLOW



Observe → Code → **Error** → **Recover**

Coding  
Harness

Runtime  
Tools

Acquire missing tools at runtime.

- ✓ Resume after unexpected errors.
- ✓ Extend with Skills & MCP.

## 02 SINGLE AGENT



last frame — next move

history-shaped

single path

Fresh reasoning. Audited evidence.

- ✓ Reopen search.
- ✓ Check the science.

## 03 TREE SEARCH



READ ONCE



BRANCH EDITS

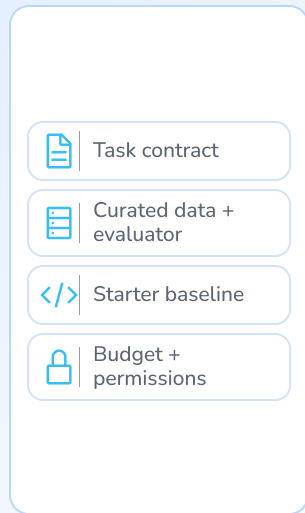
Persist the read. Vary the hypothesis.

Reuse context across experiments.

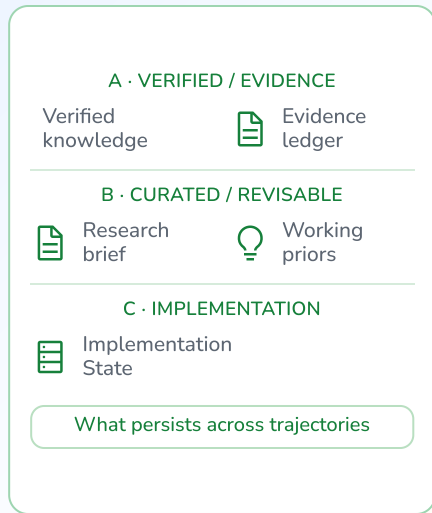
- ✓ Raise task granularity.
- ✓ Parallelize across branches.

# A system for sustained research

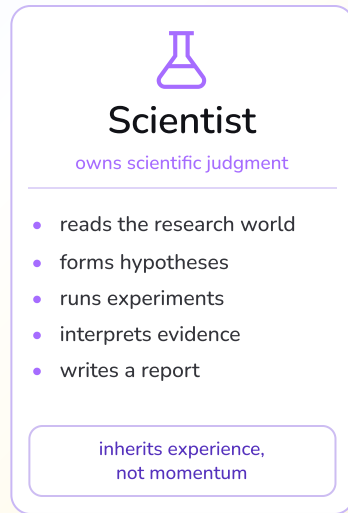
## Environment



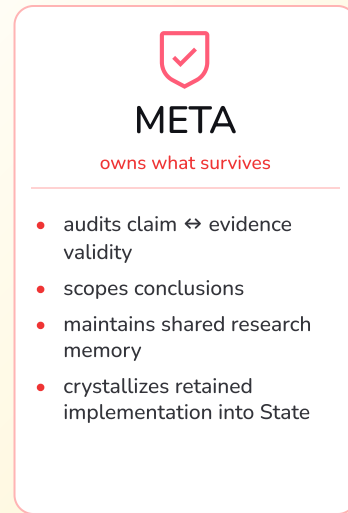
## Research World



## Fresh Scientist



## META Review



Selective persistence / trajectory handoff

## Coding-agent harness + runtime





# SciOdyssey

A research layer over your agent harness.

**SciOdyssey. A research layer over your agent harness. Your tools. One persistent research world.**

# We are not tokenmaxxing

PROJECT-LEVEL RESOURCE ACCOUNTING

## ~\$10

subscription-equivalent usage from first line of code to final result

coding · debugging · agent runs · all experiments

### MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

### COMPUTE

1 × MacBook M2

Development and experiments ran on one consumer laptop.

## Retained-run telemetry

## 48.240M

**total tokens**

including cache-read input

## 4.020M

**non-cache**

input + output

## 0

**GPU-hours**

CPU / NumPy training

# Model diversity improves the result.

Cycling GPT and Gemini brings different search styles into one research world.

## Deepen an idea

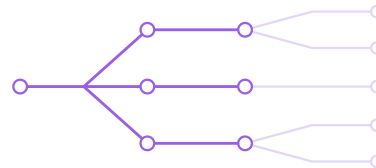
GPT 5.6 Luna



Refine the current approach.  
Follow promising details further.

## Open new directions

Gemini 3.7 Flash



Try alternative approaches.  
Challenge the current direction.

Model diversity improved the retained result.

**GPT** → **Gemini** → **GPT** → **Gemini**

🔄 Shared research memory · Gemini META

Public-validation Primary

Gemini-only

GPT + Gemini

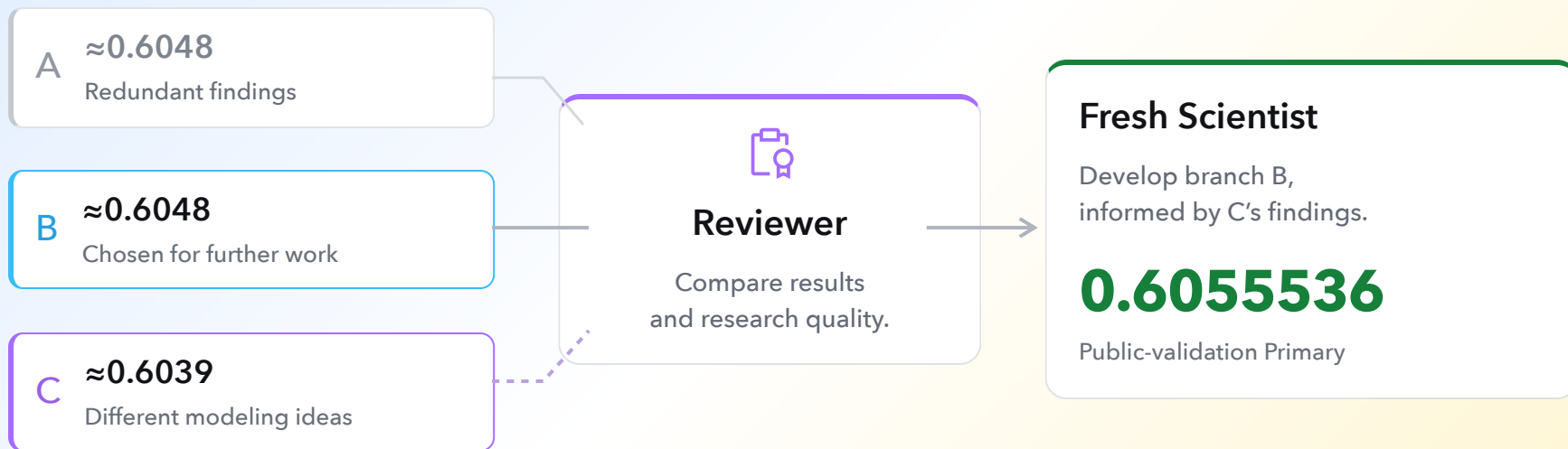
**0.6052**

**0.6059363**



# Explore in parallel. Build on the findings.

One branch continues. Another contributes a useful perspective.



A lower-scoring branch can still contribute an idea.

The new model used pairwise ranking, higher rank and rank-normalized ensembling.

# A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

## 01 MANY AGENTS



### Give every agent a branch.

**main**

clean trunk

● agent A / branch

● agent B / branch

● agent C / branch

review

→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

## 02 NO QUOTA / NO LAPTOP



### Leave the next move in GitHub.

01

Issue /  
prompt

inject the task



02

GitHub  
Action

run the agent



03

Commit

review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

Research is an **open-world** task.  
We need an **open-world** coding agent.

### 01 OPEN ACTION SPACE



Shell



Files



Browser



Git

Use what exists

### 02 SELF-RECOVERY



Read errors



Inspect docs



Inspect  
source



Install  
packages

Fix what breaks

### 03 RUNTIME EXTENSION



MCP



Skills



CLI



Packages

Add what's missing

# Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·  
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·  
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts  
· alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·  
integration support



Jiran Li

EVIDENCE /  
REPRODUCIBILITY

Telemetry · documentation  
· packaging